

Tayside Diabetes MCN Handbook

Treatment with Insulin

Insulin Preparations

	<u>Type 1</u>	<u>Type 2</u>	Time of subcutaneous injection (individual assessment recommended)
<u>Isoophane Insulin</u> (intermediate acting)			
Humulin I (Cartridge, KwikPen)		Basal insulin of choice in pregnancy	Take twice daily, about 12hrs apart
Insulatard (vial)			Take twice daily, about 12hrs apart
<u>Soluble Insulin</u> (short acting)			
Humulin S (Cartridge)			30 minutes before carbohydrate meal
Actrapid (vial)			15 - 20 minutes before carbohydrate meal
<u>Mixed Insulin</u> (short/ intermediate acting)			
Humulin M3 (Cartridge, KwikPen)		1 st line mixed insulin	30 minutes before breakfast and 30 minutes before evening meal
<u>Mixed Insulin</u> (rapid/ intermediate acting)			
Novomix 30 (Penfill, FlexPen)			Approx. 15-30 minutes before breakfast and 15-30 minutes before evening meal
Humalog Mix 25 (Cartridge, KwikPen) Humalog Mix 50 (cartridge, KwikPen)			Approx. 15-30 minutes before breakfast and 15-30 minutes before evening meal*
<u>Basal Analogue Insulin</u> (long acting)			
Semglee (glargine 100u/ml) (Pre-filled pen)	1 st line basal insulin		Once daily, but usually best before bed
Toujeo (glargine 300u/ml) (SoloStar)	2 nd line basal insulin if hypoglycaemia or requiring >1 injection per day glargine	1 st line basal insulin	Once daily
Tresiba (degludec) (cartridge 100u/ml) Tresiba (degludec) (cartridge, FlexTouch 200u/ml)	2 nd line basal insulin if hypoglycaemia or requiring >1 injection per day glargine	1 st line basal insulin for those with eGFR <60ml/min	Once daily

Prandial Analogue insulin (rapid acting)			
Trurapi (vial, cartridge, SoloStar pen)	1 st line bolus insulin		Approx. 10- 15 minutes before carbohydrate intake
Humalog (vial, cartridge, KwikPen)			Approx. 10-15 minutes before carbohydrate intake
Apidra (vial, cartridge, SoloStar)			Approx. 10-15 minutes before carbohydrate intake
Fiasp (vial, cartridge)	2 nd line bolus insulin for those who require a faster onset of action		intake Approx. 5-10 minutes before carbohydrate intake and in some individuals may be taken after eating
Highly concentrated Insulin (short acting)			
Humulin R (500u/ml) (KwikPen) (unlicensed)	To be considered in those who require >200 units of u100 insulin per day		

High strength insulin preparations/biosimilar preparation safety information

High strength licensed insulin preparations, for example Tresiba (200units/ml), Toujeo (300 units/ml) and Humalog (200 units /ml) all include risk minimisation on the eMC. The following link from the eMC takes you to Tresiba risk minimisation material - <https://www.medicines.org.uk/emc/product/7938/rmms>

The MHRA Drug Safety update (April 2015- <https://www.gov.uk/drug-safety-update/high-strength-fixed-combination-and-biosimilar-insulin-products-minimising-the-risk-of-medication-error>) provides advice for healthcare professionals